

Weihan Fei

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Education

B.S. in Artificial Intelligence, University of Science and Technology of China (USTC) Expected 2028
School of the Gifted Young Hefei, China

GPA: 3.87/4.3 Rank: 14/103

Selected Core Coursework: Mathematical Analysis (97), Discrete Mathematics (100), Probability and Mathematical Statistics (92), Linear Algebra B(1) (90), Data Structures A (90).

Self-Directed AI/CS Study: Completed full assignment tracks for Berkeley CS61B, Stanford CS229, CS230, and CS224n during Jul.–Oct. 2025, with implementations documented in GitHub repositories; studied MIT 6.S184 (Generative AI Foundations), which motivated my interest in interpretable and controllable AI systems.

Languages: Mandarin (native), English.

Research Interests

Interpretable and controllable AI systems, especially how LLMs represent, access, and revise knowledge to support reliable reasoning. I am interested in knowledge-aware reasoning, auditable agent memory, inference-time intervention, and mechanistic methods that reveal structured patterns behind model behavior.

Research Experience

Research Intern, USTC Mar 2026 – Present

Advisor: [Prof. Xiang Wang](#)

SHELF: From Similarity Retrieval to Path-Aware Auditable Memory for LLM Agents [Under Review at NeurIPS 2026]

- Reframed long-term agent memory as an interpretable knowledge-access problem: reliable reasoning requires not retrieving more similar records, but writing evidence into structural paths that future queries can revisit, verify, and diagnose.
- Co-developed SHELF, a path-aware auditable memory framework that stores facts into explicit addresses and retrieves evidence through query-conditioned routes over entity, facet, time, relation, provenance, and coverage variables.
- Designed the Auditor–Gater mechanism for variable-level path diagnosis; evaluated SHELF on LoCoMo with Qwen3-8B, Qwen3-32B, and DeepSeek-V3.2, achieving the best Overall F1, BLEU, and LLM-judge scores among compared memory baselines.

Research Intern, Alpha-Lab, USTC Nov 2025 – Present

Advisor: [Prof. An Zhang](#)

Adaptive-Thinking for Generative Recommendation

- Used generative recommendation as a controlled setting to study how explicit reasoning changes LLM-based decision-making, especially why “thinking” can improve hard cases but hurt simpler ones.
- Investigated the mechanism-level gap between “think” and “not-think” modes through history length, entropy, popularity bias, user–item affinity, and semantic item signals.
- Exploring uncertainty- and signal-aware gating strategies for adaptive reasoning, with the goal of making inference-time reasoning more selective, interpretable, and controllable.

Selected Projects

CoSwipe: AI Companion for Short-Video Feed Interaction May 2026

Third Prize, Track 1, Douyin AI Innovator Plan 2026 Hackathon, USTC Station

- Designed an AI companion experience embedded in the short-video feed, turning passive scrolling into a friend-like interaction where the AI can notice what users just watched, respond at the right moment, and make subsequent content feel more personally connected.
- Proposed three interaction styles–Mirror, Direct, and Challenge–so the companion can playfully point out behavior contrasts, name latent needs behind surface interests, and invite lightweight participation rather than only providing generic emotional comfort.

Honors & Awards

• **Undergraduate Research Opportunities Program (UROP)**, Research on Generative Recommendation Systems based on Large Language Models, advised by [Prof. An Zhang](#) Dec 2025 – Present

• **First Prize (Provincial Level, Top 20)**, The 17th Chinese Mathematics Competitions, Non-Math Major Track Oct 2025

• **Silver Prize**, Outstanding Undergraduates Scholarship Sept 2025

Skills

Research: Paper reading and synthesis, experimental design, empirical analysis, ablation studies, and end-to-end implementation for ML research.

Programming: Python, C, Java, Shell/Bash; familiar with Linux command-line workflows.

LLM Frameworks: PyTorch, Hugging Face Transformers, vLLM, verl.

Training & Inference: SFT with LoRA, RL training with verl, preference-optimization basics, inference-time prompting and reasoning.

Engineering: Linux workflows, distributed inference/debugging, experiment management, and rapid ML prototyping.